expansion stretches tiny, inhomogeneous quantum jitters and smears them clear across the sky.

Over the few billion years following the end of the brief inflationary phase, these tiny lumps continued to grow through gravitational clumping. Just as in the standard big bang picture, lumps have slightly higher gravitational pull than their surroundings, so they draw in nearby material, growing larger still. In time, the lumps grew large enough to yield the matter making up galaxies and the stars that inhabit them. Certainly, there are *numerous* steps of detail in going from a little lump to a galaxy, and many still need elucidation. But the overall framework is clear: in a quantum world, nothing is ever perfectly uniform because of the jitteriness inherent to the uncertainty principle. And, in a quantum world that experienced inflationary expansion, such nonuniformity can be stretched from the microworld to far larger scales, providing the seeds for the formation of large astrophysical bodies like galaxies.

That's the basic idea, so feel free to skip over the next paragraph. But for those who are interested, I'd like to make the discussion a bit more precise. Recall that inflationary expansion came to an end when the inflaton field's value slid down its potential energy bowl and the field relinquished all its pent-up energy and negative pressure. We described this as happening uniformly throughout space—the inflaton value here, there, and everywhere experienced the same evolution—as that's what naturally emerges from the governing equations. However, this is strictly true only if we ignore the effects of quantum mechanics. On *average,* the inflaton field value did indeed slide down the bowl, as we expect from thinking about a simple classical object like a marble rolling down an incline. But just as a frog sliding down the bowl is likely to jump and jiggle along the way, quantum mechanics tells us that the inflaton field experienced quivers and jitters. On its way down, the value may have suddenly jumped up a little bit over there or jiggled down a little bit over there. And because of this jittering, the inflaton reached the value of lowest energy at different places at slightly different moments. In turn, inflationary expansion shut off at slightly different times at different locations in space, so that the amount of spatial expansion at different locations varied slightly, giving rise to inhomogeneities—wrinkles—similar to the kind you see when the pizza maker stretches the dough a bit more in one place than another and creates a little bump. Now the normal